Pacmann

Classification Trees

Sample Dataset

- Kolom variabel → X_i
- Observasi (baris) → $(x^{(i)}, y^{(i)})$
- Label **Class** → bermain tenis/tidak

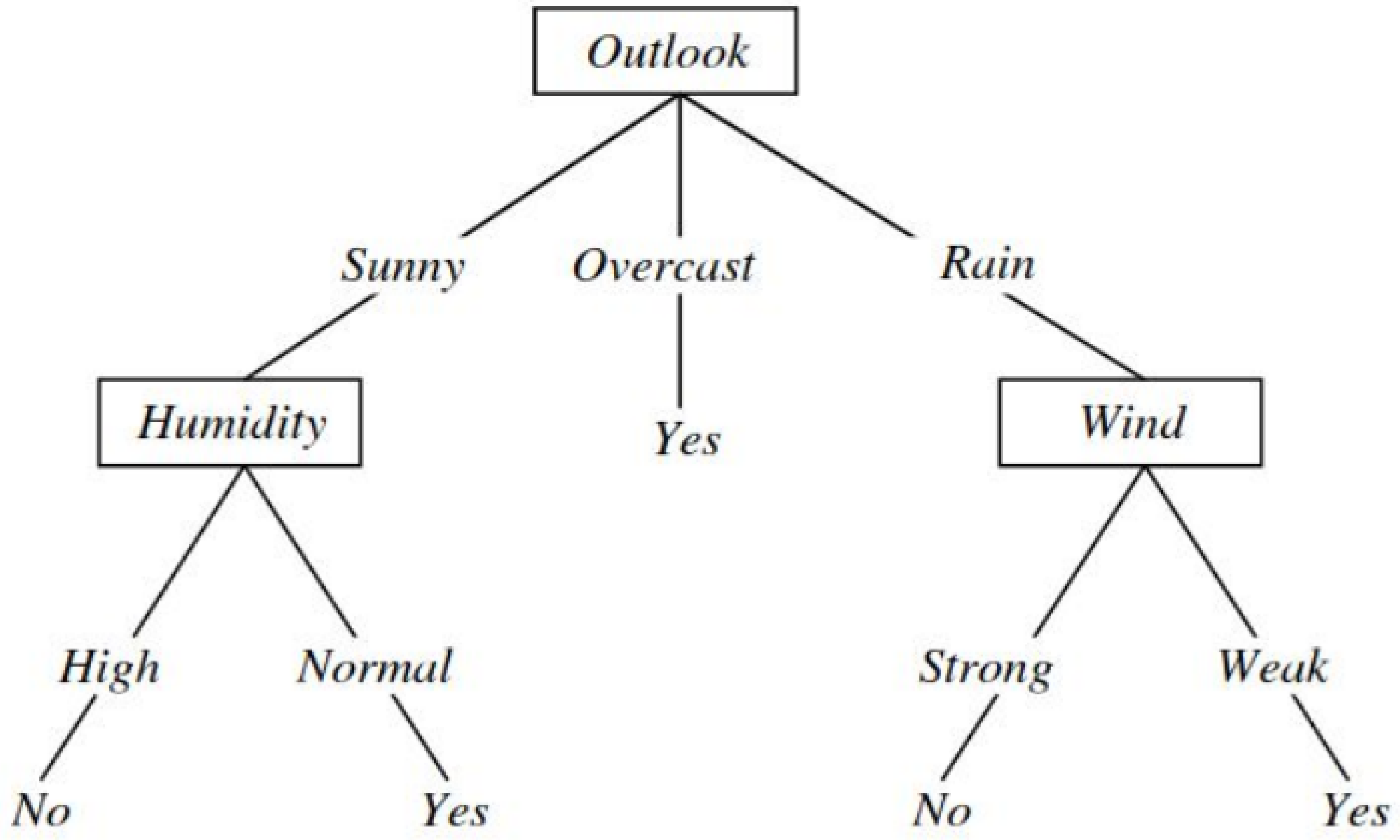
maksudnya data ke-6 x^6 $\left\langle x^{(i)}, y^{(i)} \right\rangle$

	Predictors	kelembapan	angin	Response
Outlook	**Temperature**	**Humidity**	**Wind**	**Class**
Sunny	Hot panas	High	Weak	No
Sunny	Hot	High	Strong	No
Overcast	mendung Hot	High	Weak	Yes
Rain	Mild sedang	High	Weak	Yes
Rain	Cool	Normal	Weak	Yes
Rain	Cool	Normal	Strong	No
Overcast	Cool	Normal	Strong	Yes
Sunny	Mild	High	Weak	No
Sunny	Cool	Normal	Weak	Yes
Rain	Mild	Normal	Weak	Yes
Sunny	Mild	Normal	Strong	Yes
Overcast	Mild	High	Strong	Yes
Overcast	Hot	Normal	Weak	Yes
Rain	Mild	High	Strong	No

based on slide by Tom Mitchell

Decision Tree

- Salah satu decision tree yang mungkin



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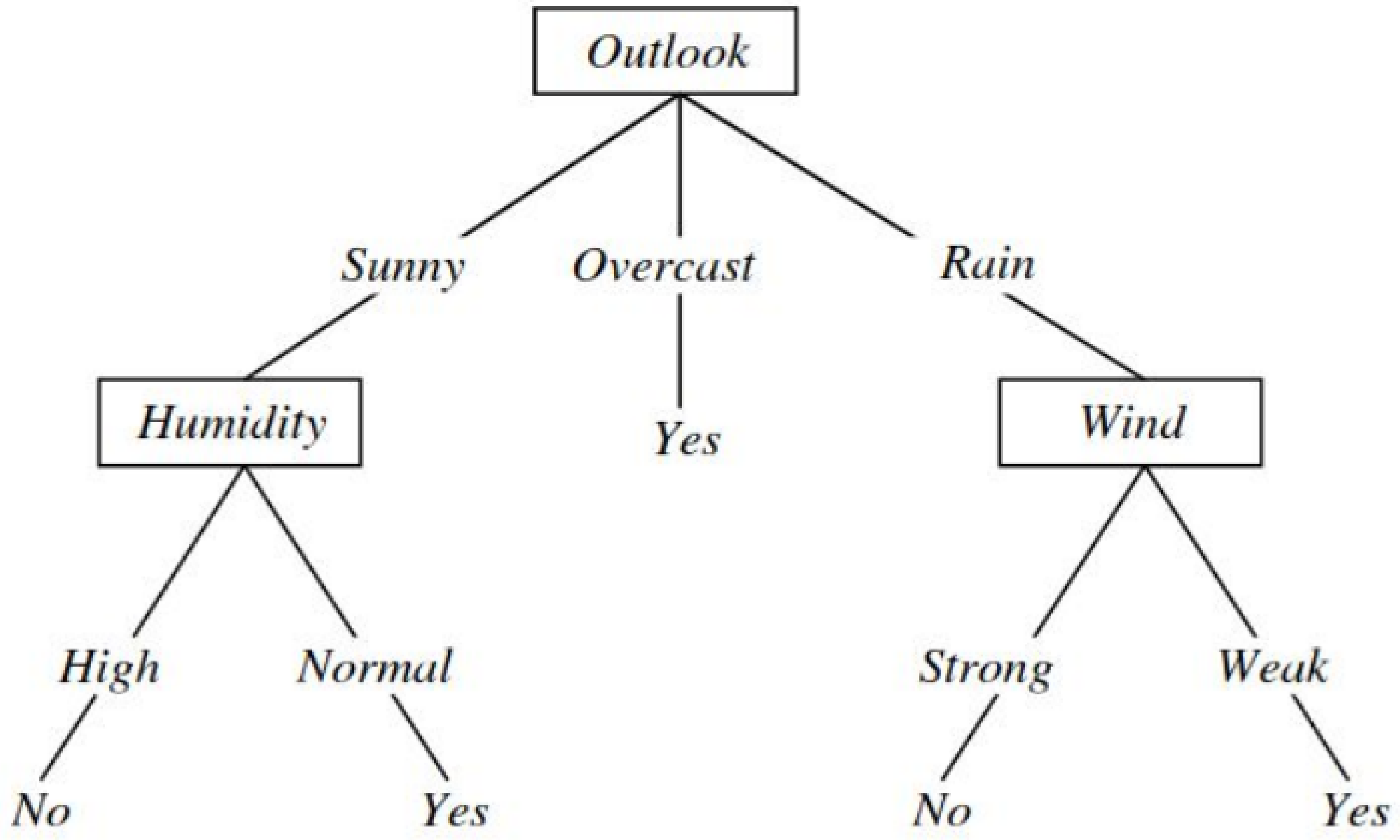
Decision Tree

- Salah satu decision tree yang mungkin
- Uji 1 atribut di tiap internal node.
- Setiap branch, pilih 1 nilai dari Xi
- Setiap leaf, prediksi Y atau, p(Y | x ∈ leaf)



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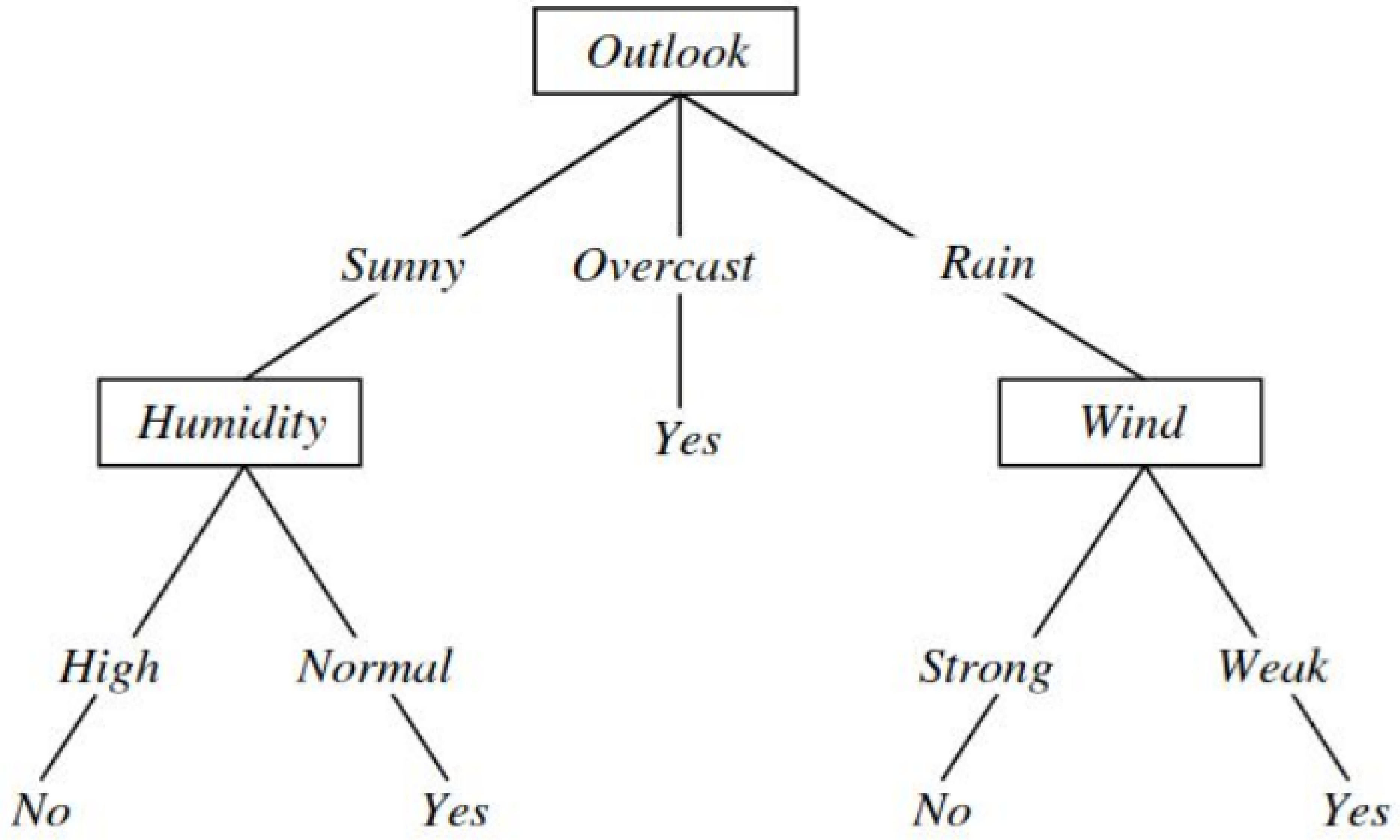
Decision Tree

- Salah satu decision tree yang mungkin
- **Bagaimana hasil prediksi** apabila:
  - outlook : sunny
  - humidity : normal



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Decision Tree

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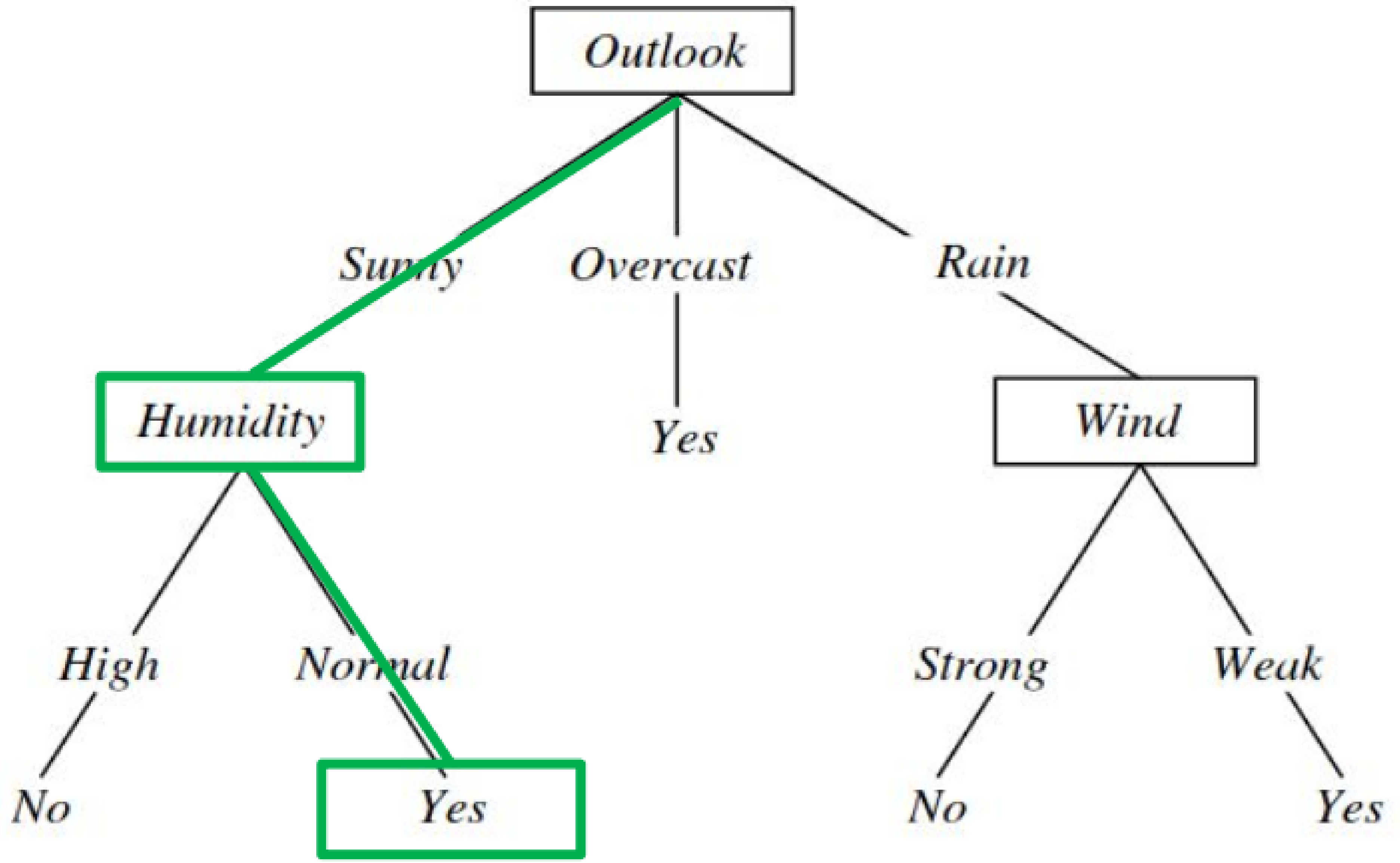

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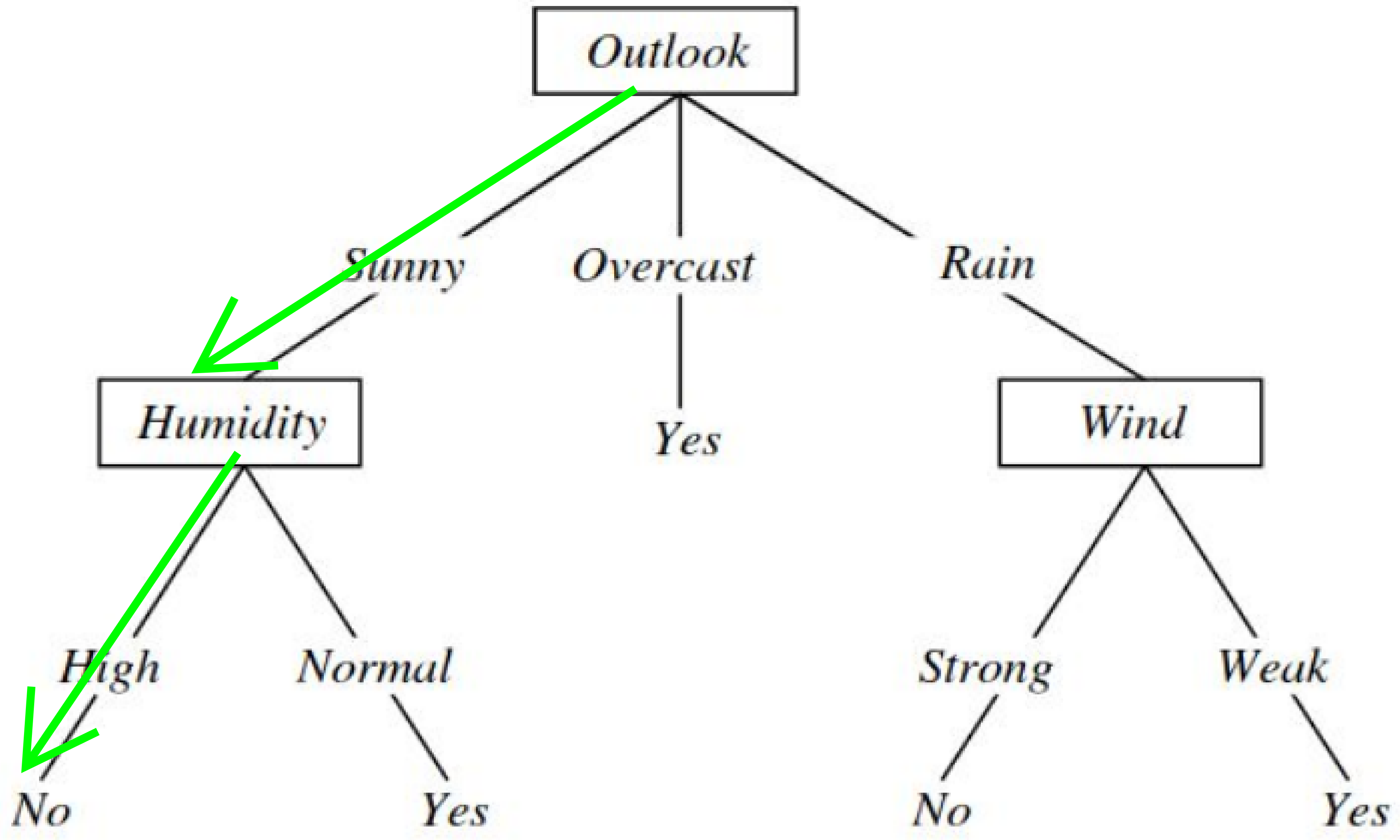
Decision Tree

- Salah satu decision tree yang mungkin

- **Bagaimana hasil prediksi** apabila:
  - outlook : sunny
  - temperature : hot
  - humidity : high
  - wind : weak



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kalo ga ada di decision tree, cukup lewatin aja, make yang ada aja

diatas karena cuman ada **outlook**
dan **humidity** doang, maka ya make 2 itu aja

temperature dan **wind ga perlu di cek lagi.**

Decision Tree

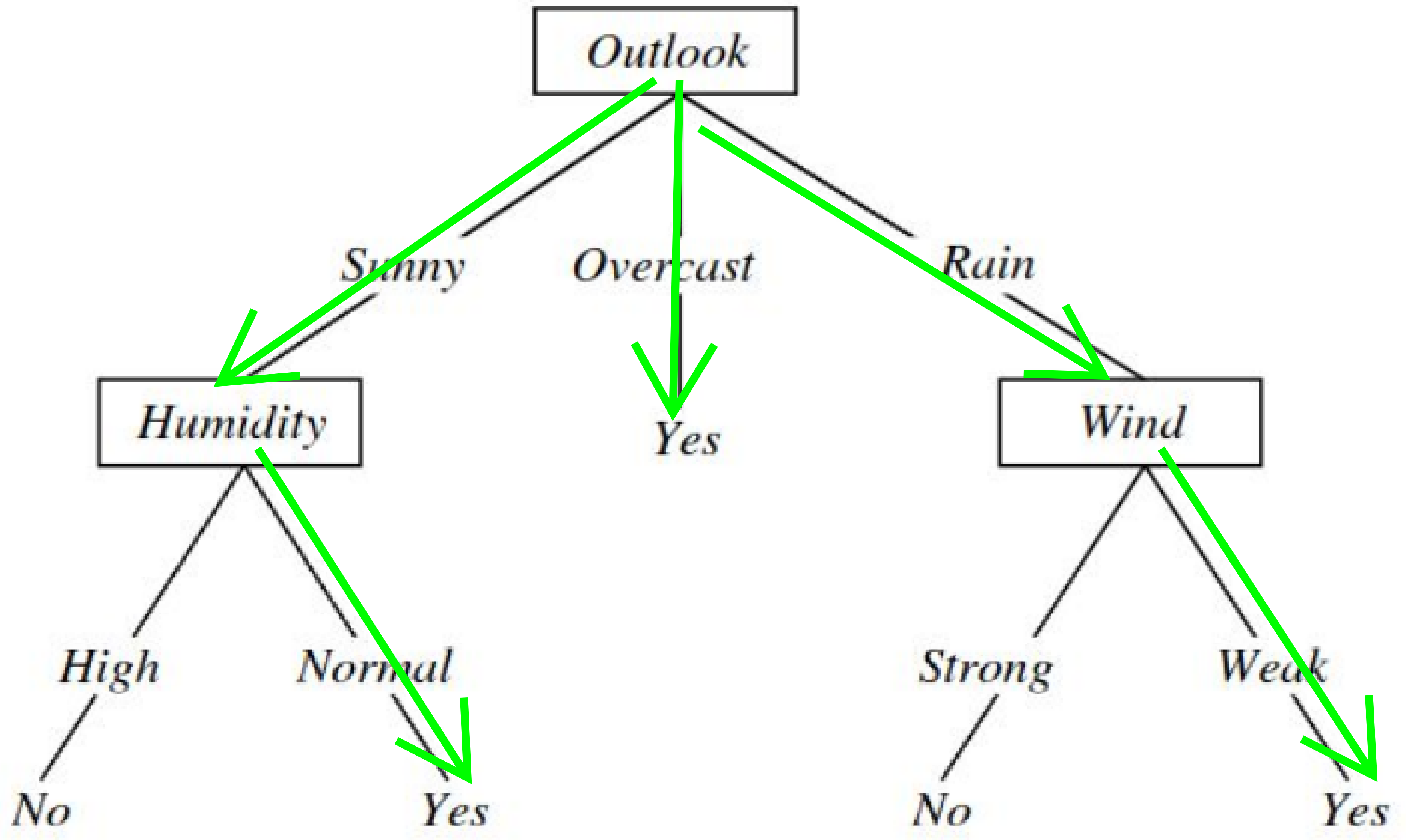
- Salah satu decision tree yang mungkin

- **Bagaimana hasil prediksi** apabila:
  - outlook : sunny
  - temperature : hot
  - humidity : high
  - wind : weak

- Hasil prediksi "yes" ketika

selain kondisi Yes di bawah,
hasilnya "**No**"

(outlook = "sunny" **dan** humidity = "normal")
atau
(outlook = "overcast")
atau
(outlook = "rain" **dan** wind = "weak")



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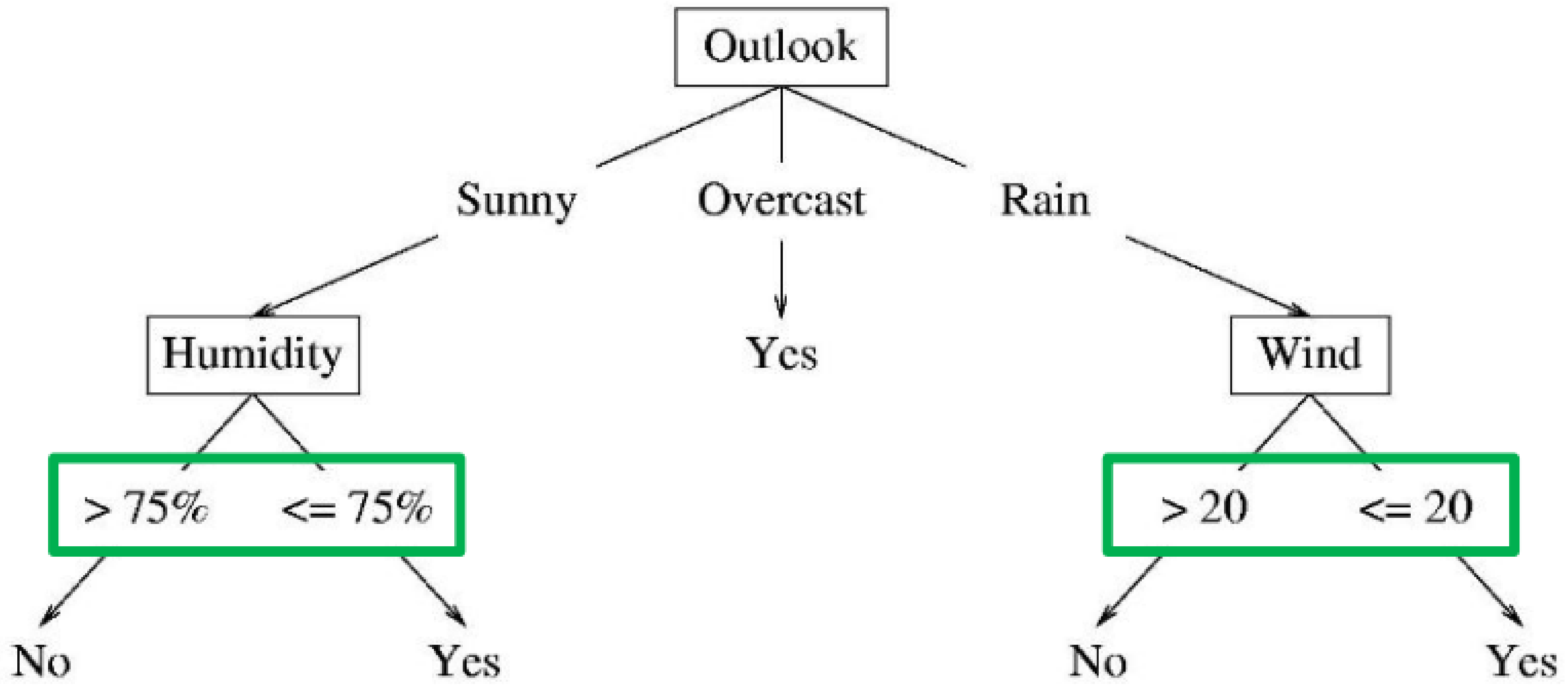
Decision Tree

- Jika fitur kontinu, **internal nodes** dapat melakukan uji nilai fitur dengan **threshold**



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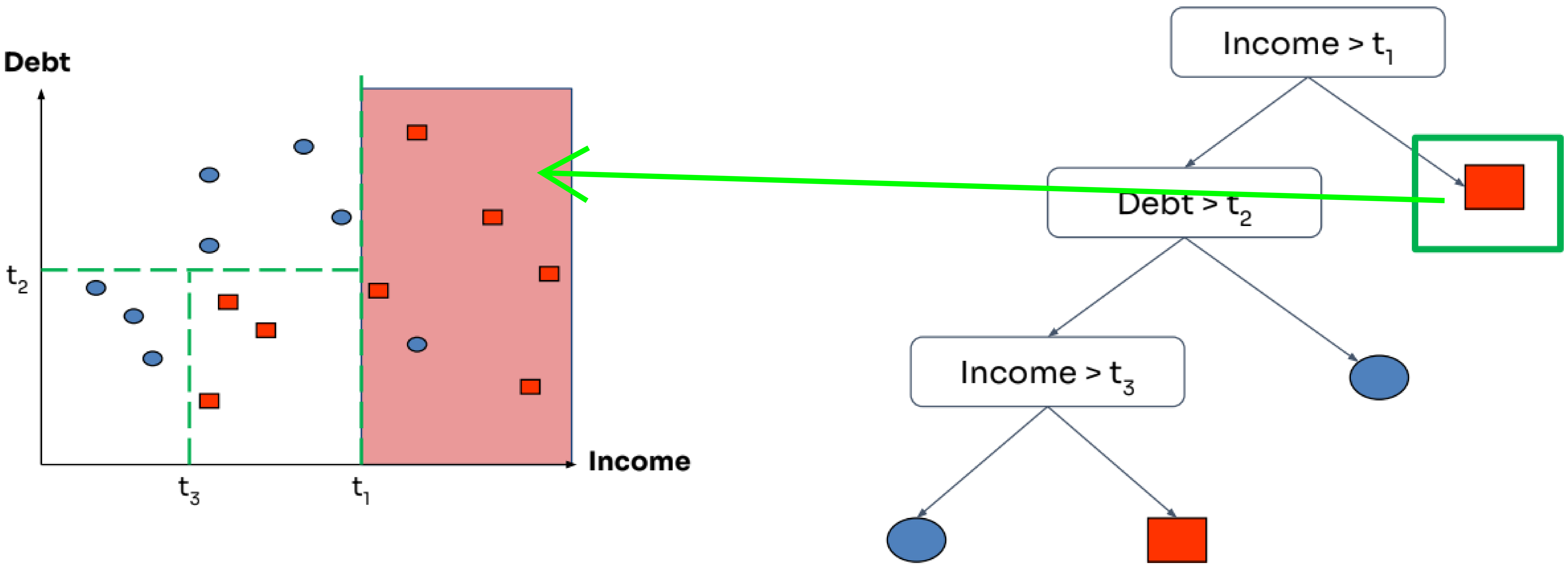
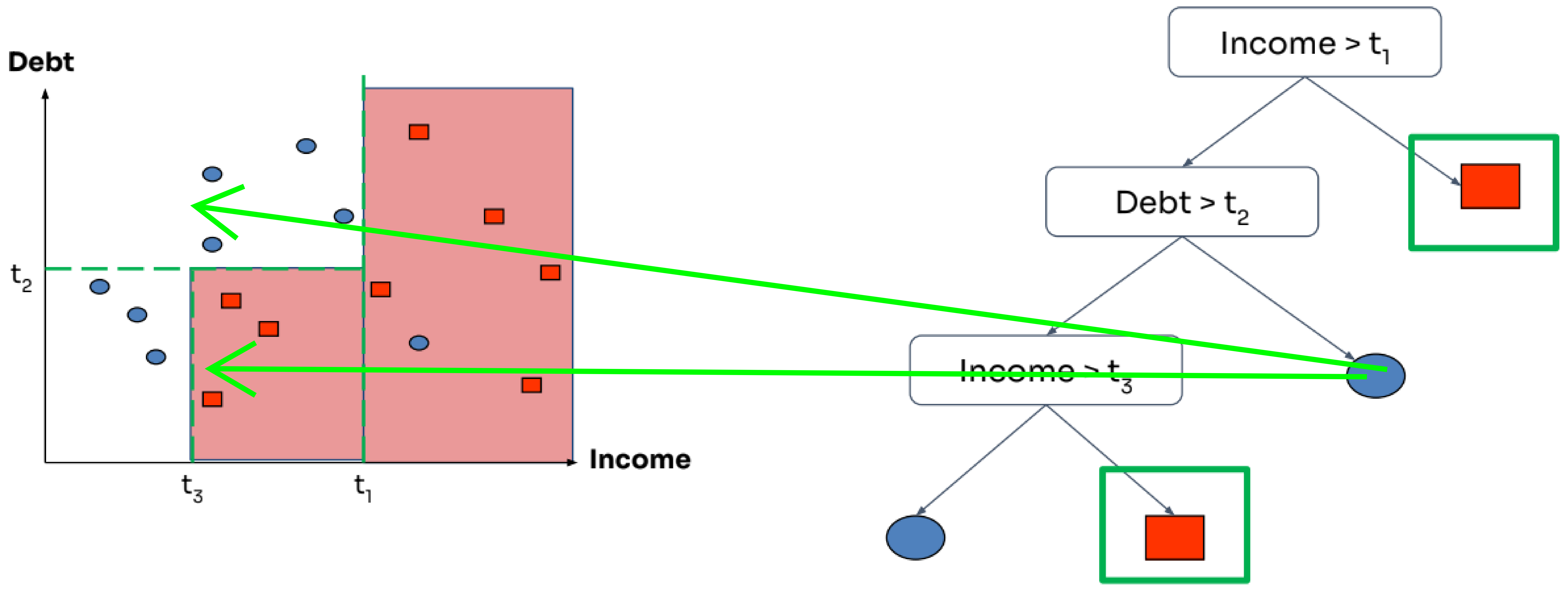
Decision Tree

Decision boundary



Decision Tree

Decision boundary



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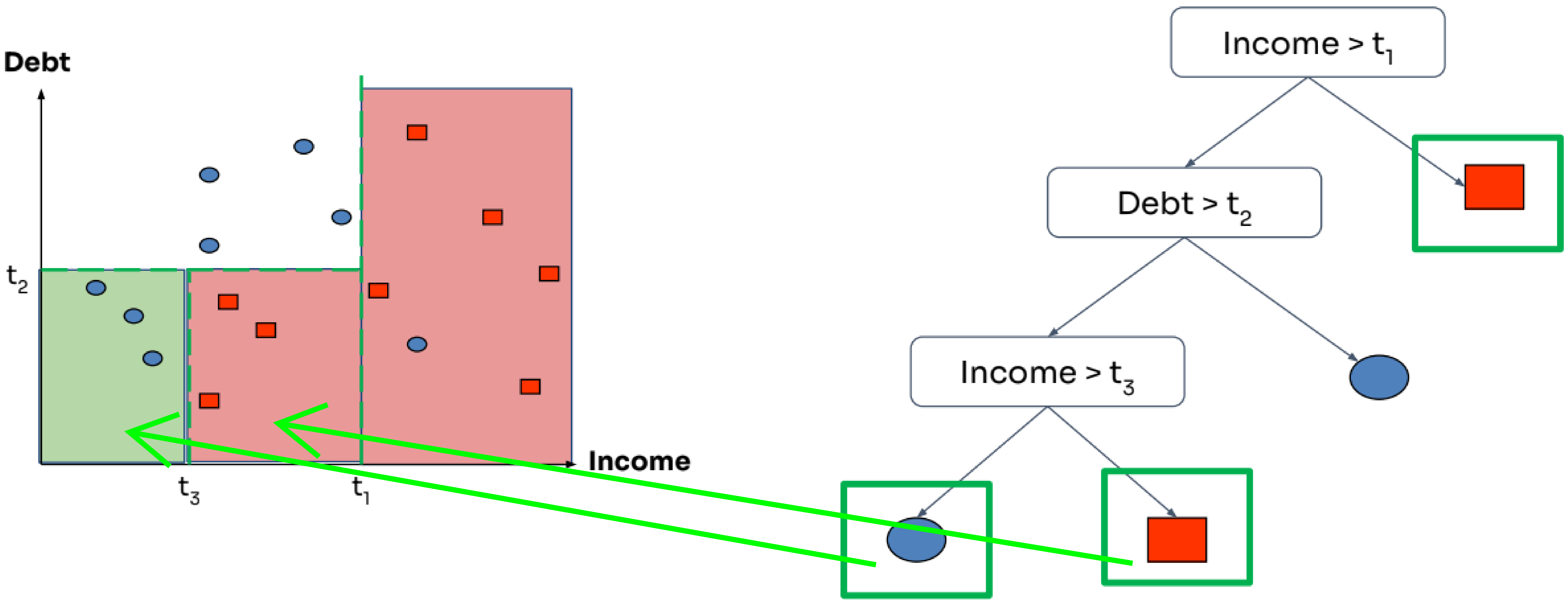
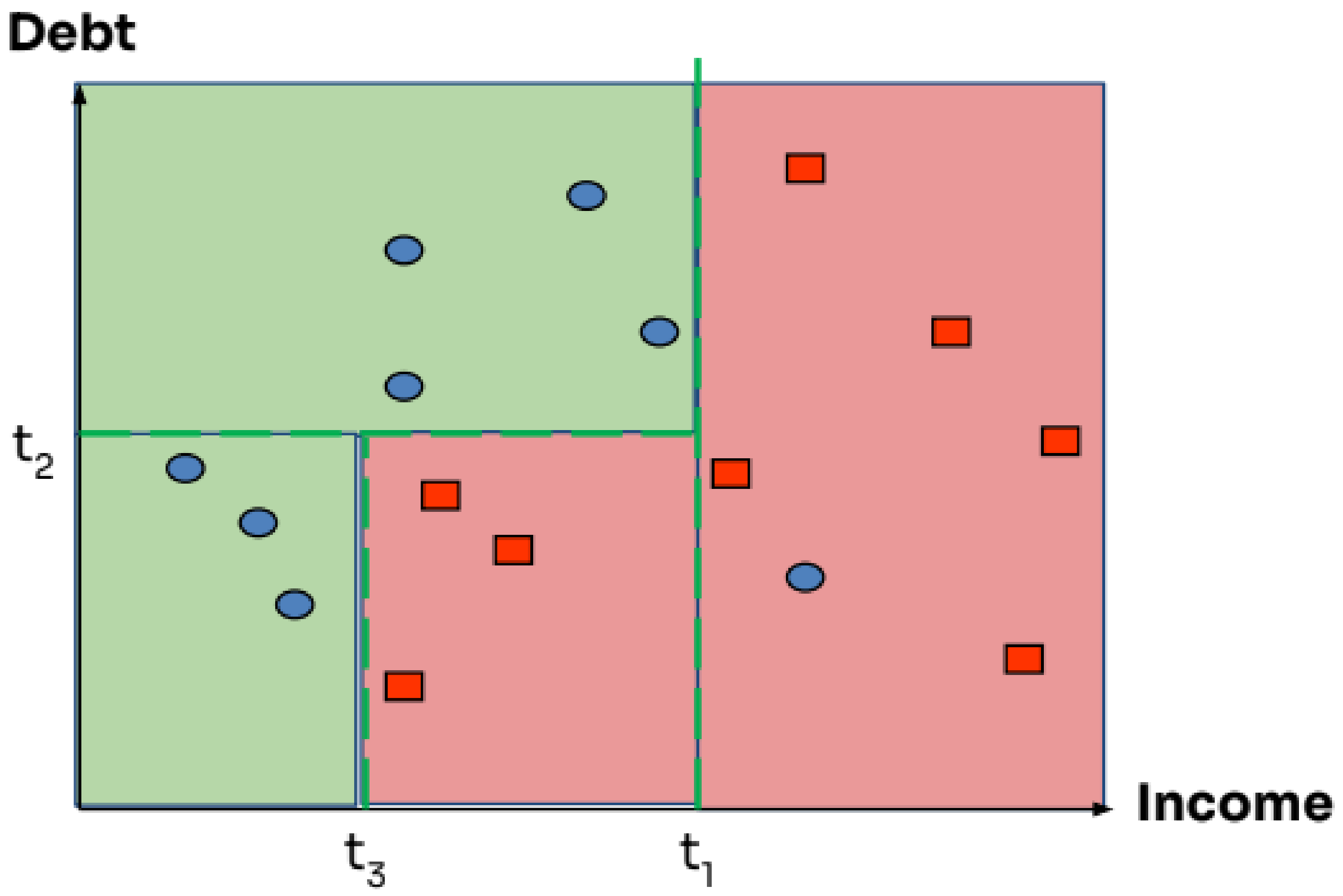
Decision Tree

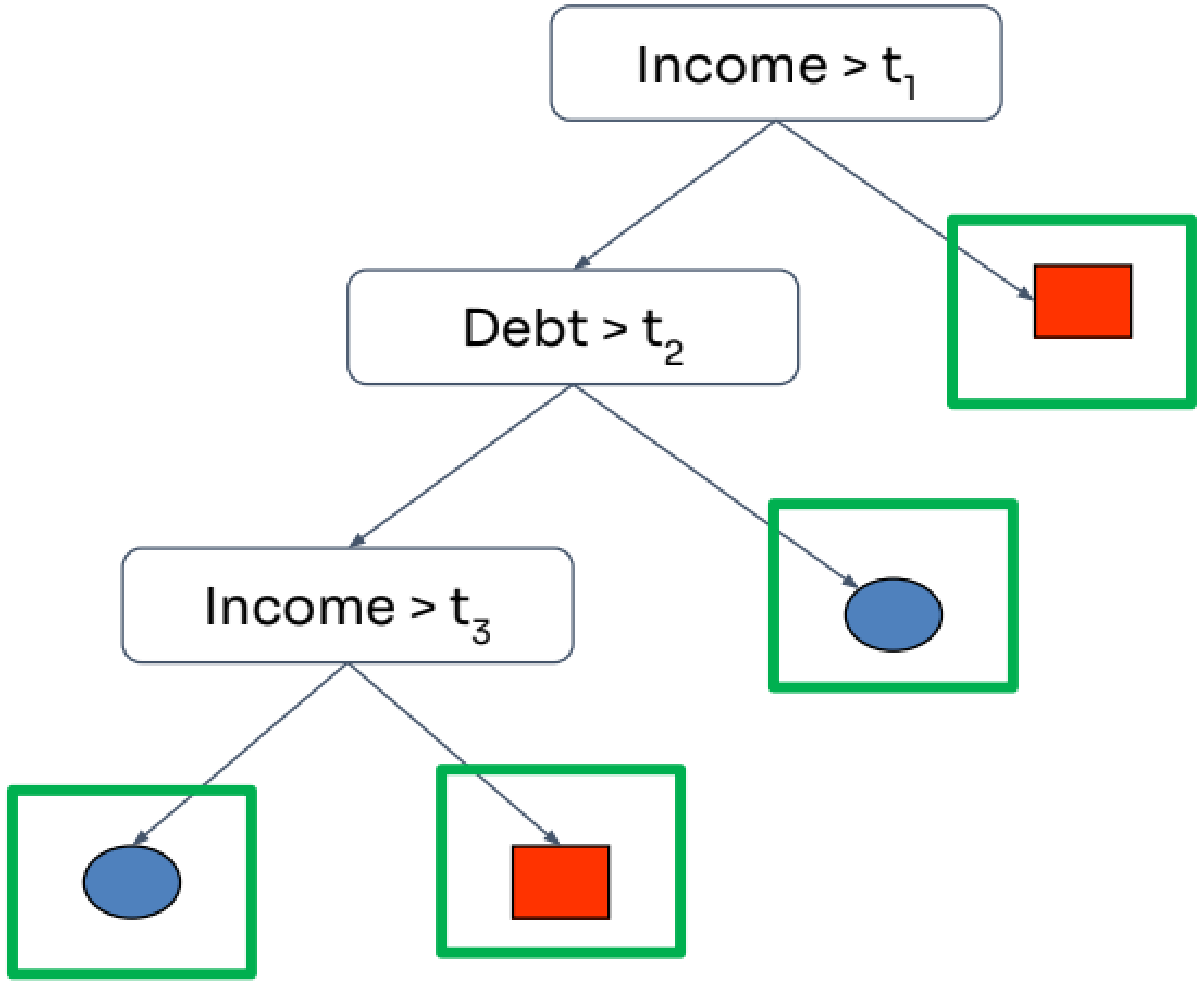
Decision boundary



Decision Tree

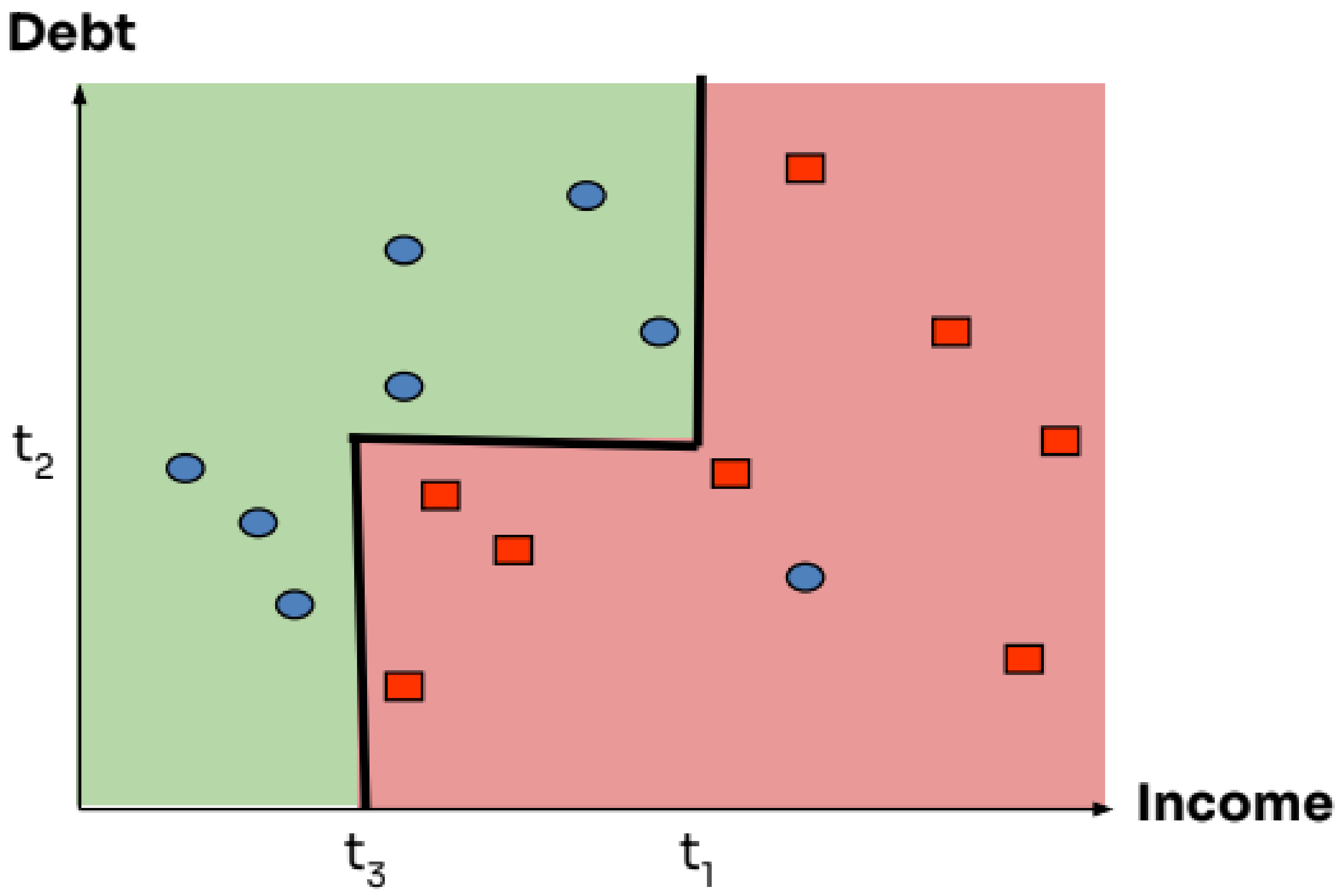
Decision boundary

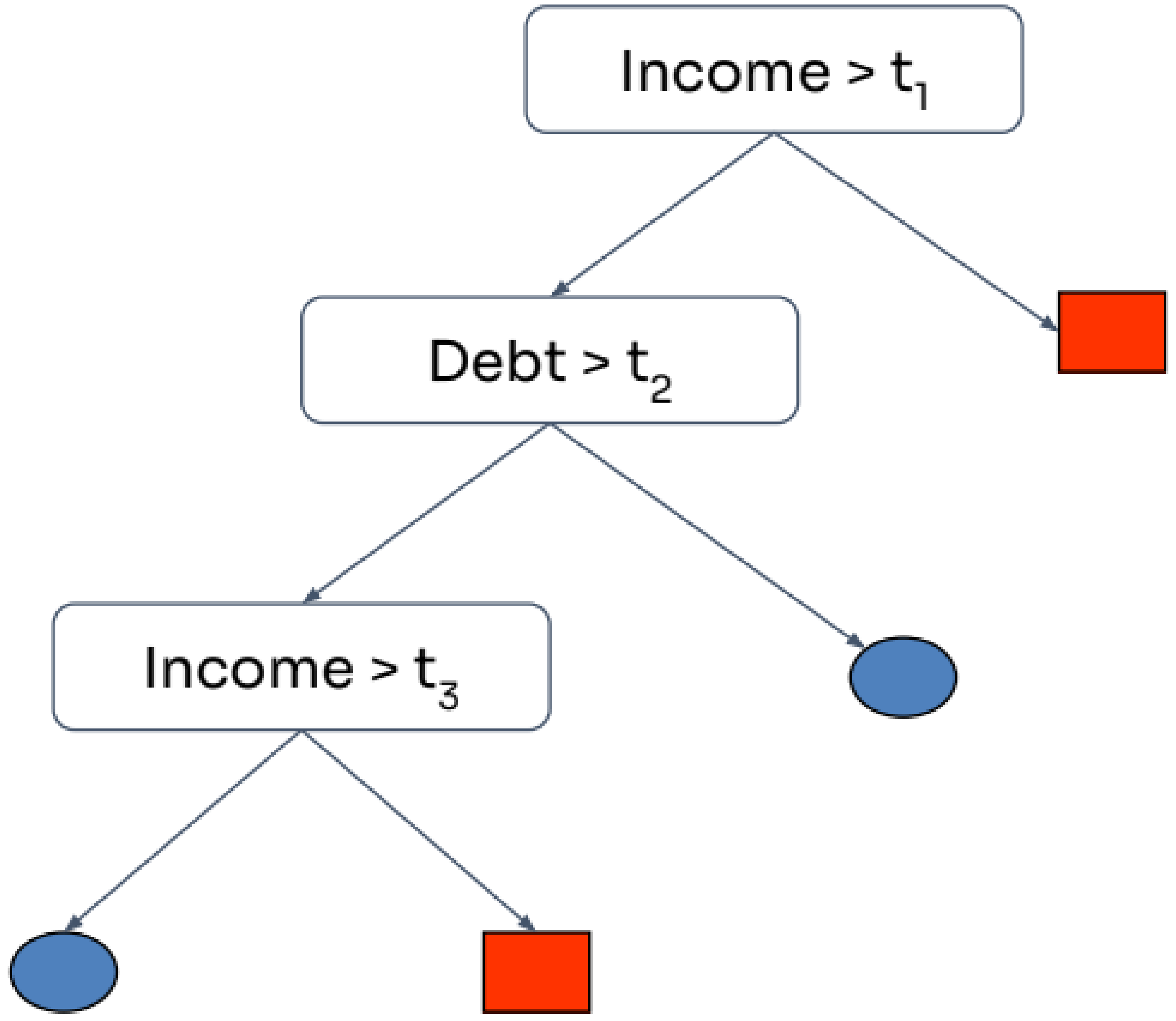




Decision Tree

Decision boundary





Decision Tree: Another Example

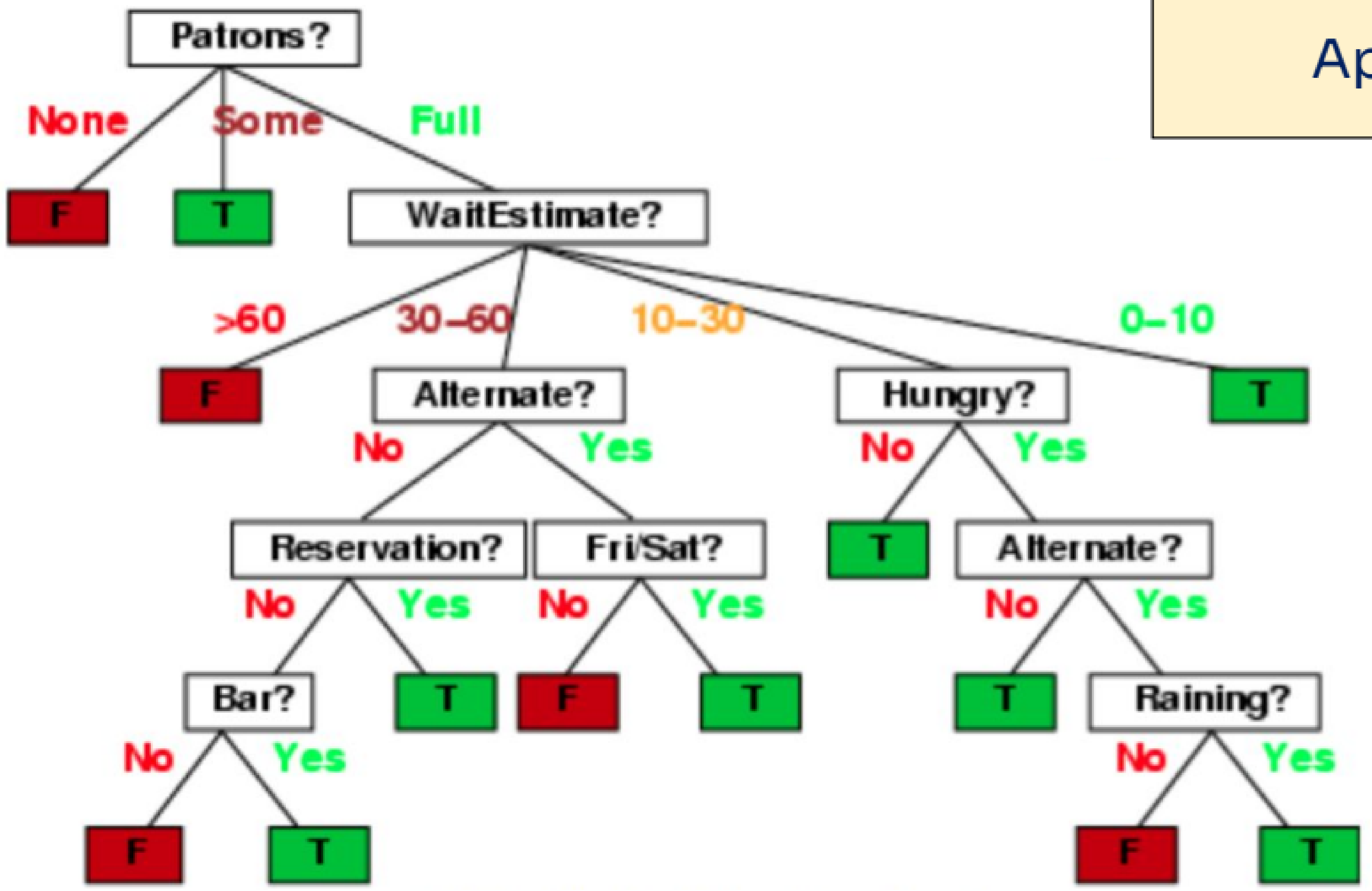
Model a patron's decision of whether to wait for a table at a restaurant

Example	Attributes										Target
	Alt	*Bar*	*Fri*	*Hun*	*Pat*	*Price*	*Rain*	*Res*	*Type*	*Est*	*Wait*
X_1	T	F	F	T	Some	\$\$\$	F	T	French	0–10	T
X_2	T	F	F	T	Full	\$	F	F	Thai	30–60	F
X_3	F	T	F	F	Some	\$	F	F	Burger	0–10	T
X_4	T	F	T	T	Full	\$	F	F	Thai	10–30	T
X_5	T	F	T	F	Full	\$\$\$	F	T	French	>60	F
X_6	F	T	F	T	Some	\$\$	T	T	Italian	0–10	T
X_7	F	T	F	F	None	\$	T	F	Burger	0–10	F
X_8	F	F	F	T	Some	\$\$	T	T	Thai	0–10	T
X_9	F	T	T	F	Full	\$	T	F	Burger	>60	F
X_{10}	T	T	T	T	Full	\$\$\$	F	T	Italian	10–30	F
X_{11}	F	F	F	F	None	\$	F	F	Thai	0–10	F
X_{12}	T	T	T	T	Full	\$	F	F	Burger	30–60	T

~7,000 possible cases

based on slide by Eric Eaton

Decision Tree: Another Example



Apa ini tree terbaik?

based on slide by Eric Eaton